Short Analysis Pipeline

BIOLM0051

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Abstract

This is a test paragraph to see how microtype improves the appearance of text. Notice how the spacing between letters and words is subtly adjusted to create a smoother, more even texture across the page. Microtype helps with justification, kerning, and overall typographic finesse, making long passages of text easier to read and giving the document a more polished, professional look.

Introduction

Methods

Some python code:

```python
from Bio import SeqIO
for record in SeqIO.parse("sequence.fasta", "fasta"):
    print(record.id, record.seq.translate())
```

Some Bash script:

```bash
#!/bin/bash
#SBATCH --job-name=blast_job
#SBATCH --partition=teach_cpu
#SBATCH --time=01:00:00
#SBATCH --mem=2000M

blastn -db nt -query sequence.fa -out results.out -remote
```

Results

In this study, we analysed the gene expression levels across multiple samples using RNA-seq data. The expression values x_i for gene i were normalised using the log-transformation:

$$y_i = \log_2(x_i + 1)$$

to stabilise variance. Data processing was performed using Python scripts:

```
import pandas as pd
data = pd.read_csv("expression.csv")
normalized = data.apply(lambda x: np.log2(x+1))
```

Discussion

Yadadadada [1] and then also [2] and maybe even from this guy [3, 2]

References

References

[1] WJ Kent. Blat—the blast-like alignment tool. *Genome research*, 12(4):656–664, 2002.

[2] MJ Wolin, TL Miller, and CS Stewart. Microbe-microbe interactions. In *The rumen microbial ecosystem*, pages 467–491. Springer, 1997.

[3] B Baker, P Zambryski, B Staskawicz, and SP Dinesh-Kumar. Signaling in plant-microbe interactions. *Science*, 276(5313):726–733, 1997.

Supplementary Data

Please find supplementary data within the GitHub repository for this project: https://github.com/archiebenn/intro_to_bioinformatics_report.git